

Exercises 1.6

1. Find the argument form for the following argument and determine whether it is valid. Can we conclude that the conclusion is true if the premises are true?

If Socrates is human, then Socrates is mortal.

Socrates is human.

∴ Socrates is mortal.

Soln: p : Socrates is human q : Socrates is mortal

The argument is of the form:

$$\frac{p \rightarrow q}{p}$$

This is modus ponens, which is a valid argument. ∴ The conclusion is true, since the hypotheses are true.

2. Find the argument form for the following argument and determine whether it is valid. Can we conclude that the conclusion is true if the premises are true?

If George does not have eight legs, then he is not a spider.

George is a spider.

∴ George has eight legs.

Soln: p : George is a spider q : George has eight legs

The argument is of the form:

$\neg q \rightarrow \neg p$
p
∴ q

By double negation law,
 $\neg(\neg p) \equiv p$ and $\neg(\neg q) \equiv q$

$$\frac{\neg q \rightarrow \neg p}{\neg(\neg p)}$$

$$\therefore \neg(\neg q)$$

This is modus tollens, which is a valid argument. ∴ If the premises are true, then the conclusion is true. $\square$